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Branding is what people say about you when you are not in the room.

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-Jeff Bezos



based on volume. For line balancing, each stage of manufacturing needs to be designed so that the product moves along the assembly line at the right speed to optimise the flow.

From a test perspective, there are usually two stages in a typical assembly line other than an offline QA. While the PCB gets populated with components, it needs to be tested for open/short, for which a bed of nails system is used. Then, if there are components/circuits which need calibration, a test system is needed for this purpose.

A case in point is mobile manufacturing where the RF board needs to be calibrated for power and frequency. Once this is done, at the final stage of assembly, the product needs to go through a functional test before it can be packaged and shipped to customer. Both these test systems need to be carefully designed and optimised so that quality and cost are optimised.

For design stages, customers usually perform many tests in manual mode, but at the end of design cycle they go through repeatability and reproducibility (R&R), which is a good candidate for a test automation where usually the complete suite of tests is performed. Here, test time reduction is not critical, more important is coverage of all tests and uploading of test data to database.

A critical aspect of a test system is when the DUT is in PCB form and the connectivity to the test system is through a fixture. The fixture design should be rugged and precise, since the test points in the PCBs must be contacted precisely to avoid any shorting of adjacent connections. Once the connectivity is established, the test software sequences each test based on the testplan.

If the DUT is assembled, it needs to go for a functional test defined by the design team. Here, the test

system needs to provide power to DUT, provide all electrical interfaces (DC, digital, RF), and run the test plan that programs various instruments in the test rack to provide the needed test conditions to the DUT.

Sometimes, the DUT itself needs to be programmed through a digital interface (USB, serial, custom digital IO) that puts it in the test mode. Then the tests are run as per pre-defined sequence, test data collected and analysed, and decided whether it passed or failed.

The optimum design of a test system, integration, software development, and ensuring it meets end customer's needs requires a dedicated team with a system engineer to understand customer's test requirements and a project manager to manage the lifecycle of the project. Test equipment vendors and their channel partners are gearing up their manpower infrastructure to cater to this need and adding the required investment.

The test system which solves a customer's real need has many components—test instruments, cables and connectors, fixture, rack, PC with customised software. A single vendor, either OEM or a channel partner, will not have all these components and may have to source from various sources.

End customers have also matured to choose vendors who have this solution design and delivery capability instead of bits and pieces. A vendor providing a complete solution takes responsibility to ensure end customer's goals are met—a complete test system with mutually agreed deliverables like testplan, optimised test time, connecting test data to ERP, support phase, etc.

There are many test and measurement vendors and their distributors who have felt this pulse of the customers about their need for a total solution and are adding this capability. **EFY**